

A RESEARCH ON FACTORS THAT MAY INFLUENCE INFANT'S BIRTH WEIGHT

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Introduction

Low birth weight (LBW) is defined as an infant weighing less than 2.5 kg at birth, regardless of gestational age (Elliza, Norliza, Nor Afiah, Ying Lim, 2024). It is one of the common issue faced during pregnancy outcome. There are multiple short term and long term health issues that the baby might have due to the low birth weight including predisposed to fetal complications such as neonatal respiratory distress syndrome and certain congenital abnormalities and experience issues with growth and development to name a few. For pregnant mothers, the birth of LBW infants can negatively impact their mental health, leading to anxiety and depression (Elliza, Norliza, Nor Afiah, Ying Lim, 2024). Our aim is to investigate on whether the age of the mother and gestational period affect the infant's weight at birth.

Research Question

Is an infant's birth weight influenced by the mother's age and gestational period?

Research Objectives

- To determine the relationship between a mother's age and the birth weight of infants.
- To examine the association between gestational period and infant birth weight.
- To analyze the combined influence of mother's age and gestational period on infant birth weight.
- To quantify the strength and direction of the effects of maternal age and gestational period on infant birth weight.

Importance of the Study

- To educate people on factors that may influence the infant's birth weight at birth
- To give awareness towards the couples who planned to have a baby.
- To provide insights to the practitioners and policymakers in the related field (Obstetric and Gynaecology).

Data Collection

For this research, data were collected from a sample of 30 individuals measuring three different variables for analysis. The variables are:

- Age : Mother's age during pregnancy
- Gestational_period: Number of days baby develop in the womb until the baby is born
- Newborn_weight: The infant's weight at birth

For gestational period, the original data collected was gestational weeks plus days (37 weeks + 6 days = 265 days). However, for the purpose of this research, gestational period is measured in days for simplicity.

Exploratory Data Analysis

Summary of descriptive statistics of all the three variables are shown in below table:

Variables	Min	1 st Quartile	Median	Mean	3 rd Quartile	Max	Std Dev	Var	CV (%)	Skewness	Kurtosis
Mother's Age	22	28	31.5	31.0333333	34	39	4.5142176	20.3781609	14.5463512	-0.3036868	-0.8363601
Gestational Period (days)	252	265.25	269	268.6666667	273	280	6.5932742	43.4712644	2.4540723	-0.4189342	0.1504722
Newborn Weight (kg)	2.3646	2.87455	3.00975	3.0350667	3.17475	3.775	0.3236675	0.1047606	10.6642629	0.3099325	-0.1265888

Coefficient of Variation (%)

- Mother's Age has the highest CV compare to other two variables. However, it is still within a moderate variability towards the mean.

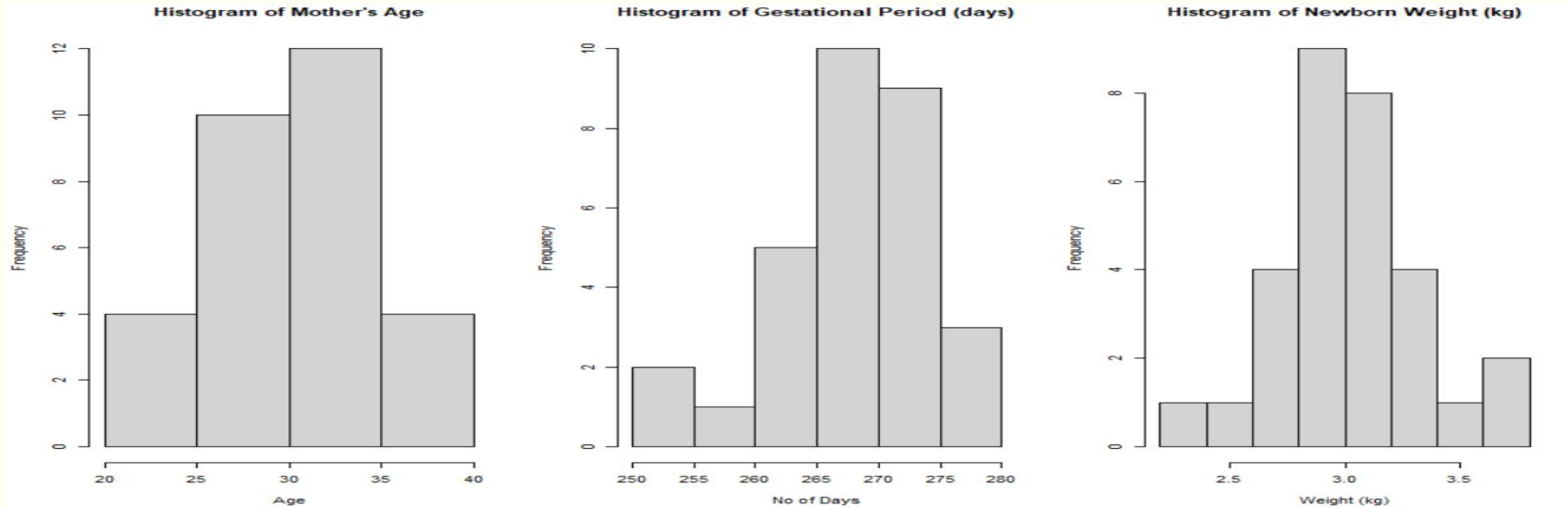
Skewness

- Both Mother's Age and Gestational Period has negative skewness value (left skewed; mean<median). Meanwhile, Newborn weight has positive skewness value (right skewed; mean>median).

Kurtosis

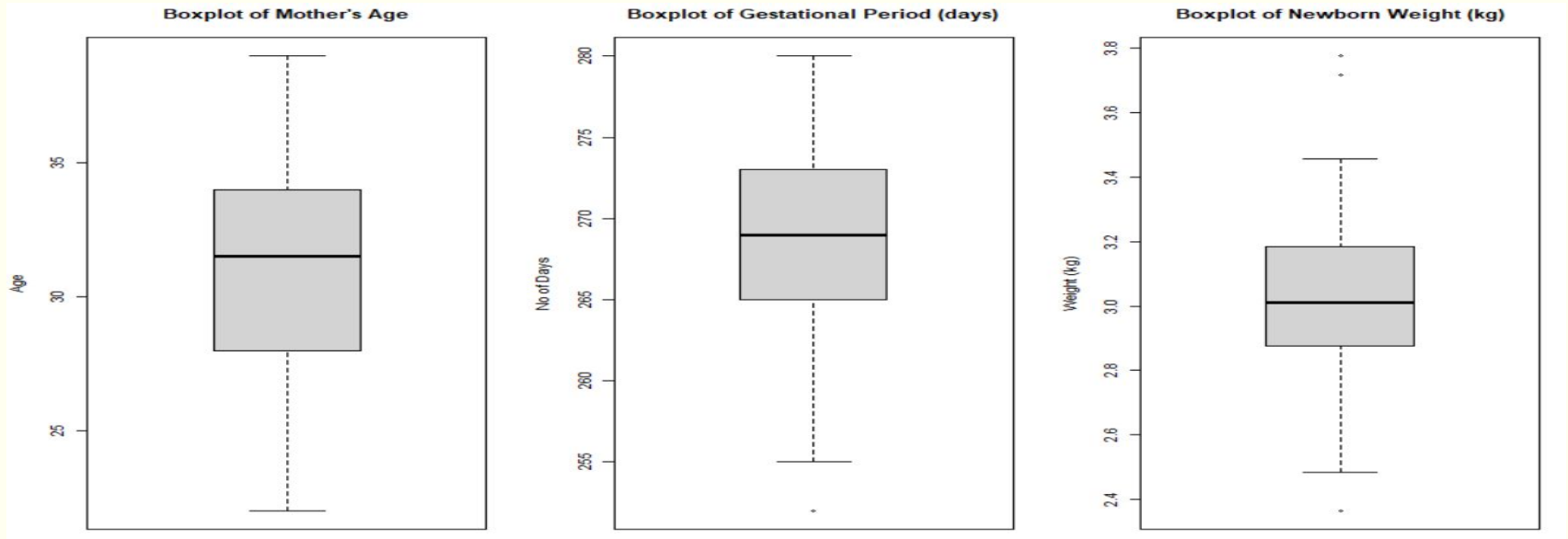
- Both Mother's Age and Newborn Weight have negative kurtosis value (platykurtic); where the distributions have short tails. However, Gestational Period has positive kurtosis value (leptokurtic); where the distributions have longer tails.

Histograms for all three variables



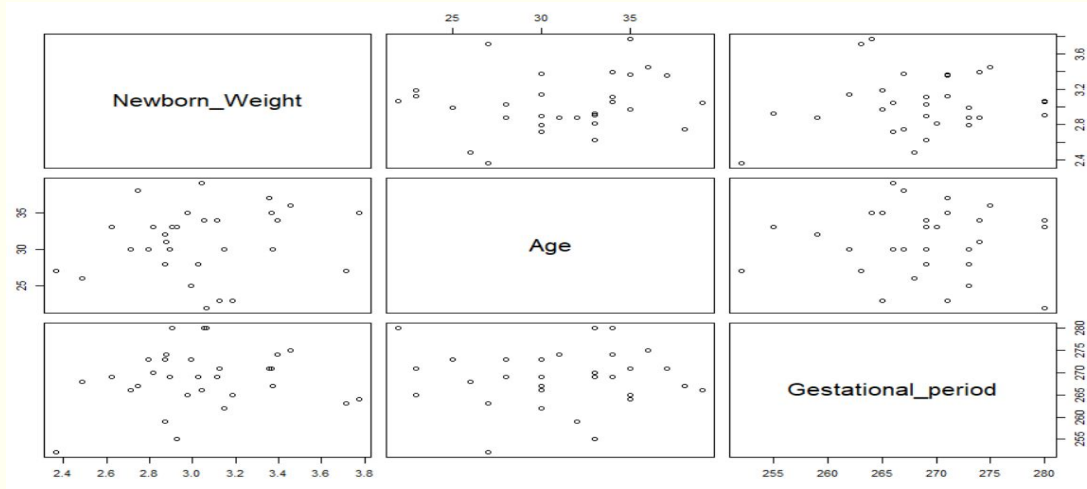
From the histogram plots above, all three variables seem to have a normal distribution around the mean value. However, as per the table previously, the distribution is not exactly normal as per the skewness values suggested that Mother's Age and Gestational Period have left skewed distribution and Newborn Weight has right skewed distribution.

Boxplots for all three variables



From the boxplots above, there are no outlier for Mother's Age (fairly spread). However, there is an outlier visible for Gestational Period at the bottom part of the plot. Newborn Weight plot visualizes 3 outliers with 2 of them are above the maximum value and one of them below the minimum value which is something to take into consideration as it falls under the Low Birth Weight.

Linear Regression



Scatterplot matrix above indicates that there is no clear pattern between each of the variables towards each other.

The fitted model (based on the collected data) of dependent variable; Newborn_Weight (y) against independent variables; Age (x_1) and Gestational_period (x_2) is shown below:

$$\text{Newborn_Weight} = 0.701199 + 0.012083\text{Age} + 0.007291\text{Gestational_period}$$

- 0.012083 value means an increase of mother's age by 1 unit will increase the newborn weight by 0.012083kg considering Gestational_period remain constant.
- 0.007291 value means an increase of Gestational_period by 1 day will increase the newborn weight by 0.007291kg considering Age remain constant.
- However, the adjusted R-squared value for this fitted model is negative (-0.0201) which suggests that the model's explanatory variables (Age & Gestational_period) do not have a meaningful relationship with the dependent variable.
- Hypothesis testing must be performed in other to find out the significance of the independent variables towards the dependent variable.

Hypothesis Testing

Based on the fitted models, the hypotheses are

$$H_0: \beta_1 = \beta_2 = 0$$

$$H_1: \beta_j \neq 0 \text{ for at least one } \beta_j, \text{ where } j = 1, 2$$

Where,

- β_1 is Age
- β_2 is Gestational_period

The F-statistic, F_0 based on the fitted model is 0.7142. The critical value $F_{0.05, 2, 27}$ at 95% significance level ($\alpha = 0.05$) is ≈ 3.35 . Since $F_0 < F_{0.05, 2, 27} = 3.35$, we fail to reject H_0 . Hence, there is no enough evidence to claim that β_1 (Age) and β_2 (Gestational_period) are not equal to zero. Therefore, the conclusion is that Age and Gestational_period does not have linear relationship with Newborn_Weight.

The p-value of each variable are

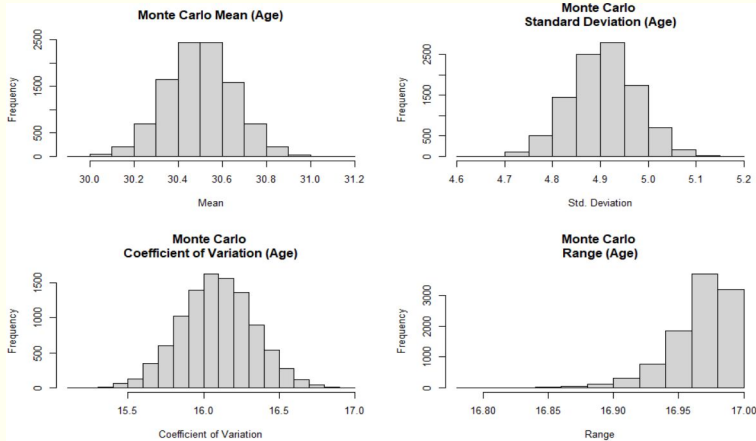
- Age = 0.377
- Gestational_period = 0.435

Compare this at 95% significance level ($\alpha = 0.05$), all p-values are above 0.05 (>0.05) which indicates that the null hypothesis is failed to reject. Hence, conclude that Age and Gestational_period individually does not have linear relationship with Newborn_Weight. Based on the research objectives, the conclusion is that there is no relationship between a mother's age and the birth weight of infants, between gestational period and infant birth weight as well as no combined influence of mother's age and gestational period towards infant birth weight.

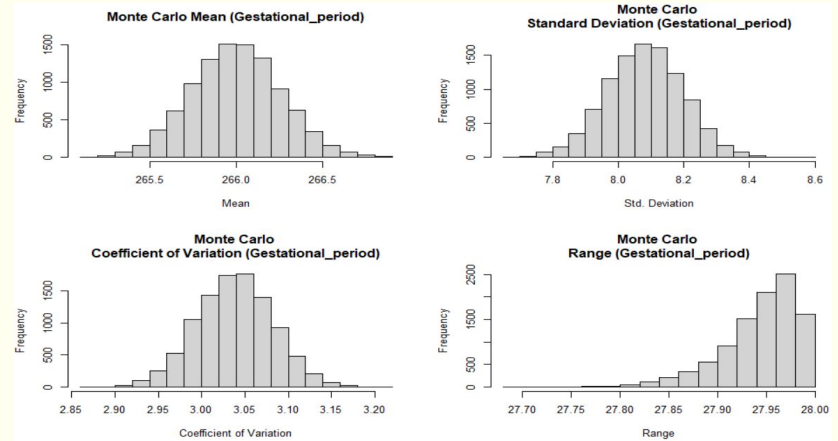
Monte Carlo

For each variable, 10000 Monte Carlo samples of size 1000 are generated and the mean, standard deviation, coefficient of variation and range for each sample are estimated. Sampling distributions for each variable are shown below:

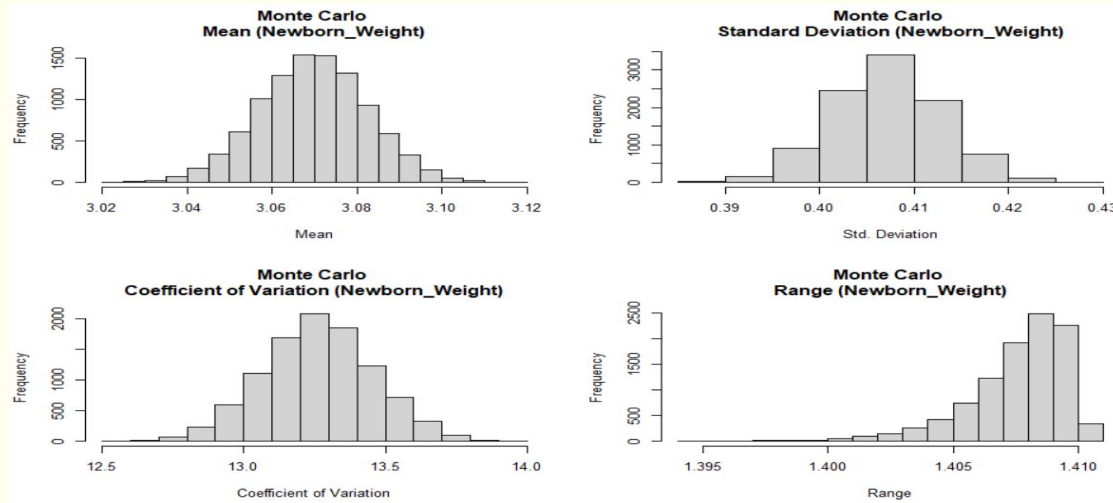
- Age (uniform distribution assumption)



- Gestational_period (uniform distribution)



- Newborn_Weight (uniform distribution assumption)



For all three variables, mean, standard deviation and coefficient of variation have normal sampling distributions. However, the range sampling distribution for all variables skewed to the left. This indicate that the range is more sensitive to the extreme values. This is because just one extremely high maximum or low minimum value can drastically change the range.

Conclusion

In conclusion, mother's age and gestational period does not influence the infant's weight at birth. There may be other external or internal factors that need to be taken into account when considering variables that can affect the infant's birth weight. Factors such as maternal nutritional status, lifestyle behaviors (smoking or alcohol use), and medical conditions (hypertension or anemia) have been shown in other research to influence birth weight outcomes. Further research needs to be done to discover more on this matter.

Reference

- Elliza Mansor, Norliza Ahmad, Nor Afiah Mohd Zulkefli, Poh Ying Lim (2024) **Incidence and determinants of low birth weight in Peninsular Malaysia: A multicentre prospective cohort study**